

DRONES, UAVS AND ROBOTICS POLICY

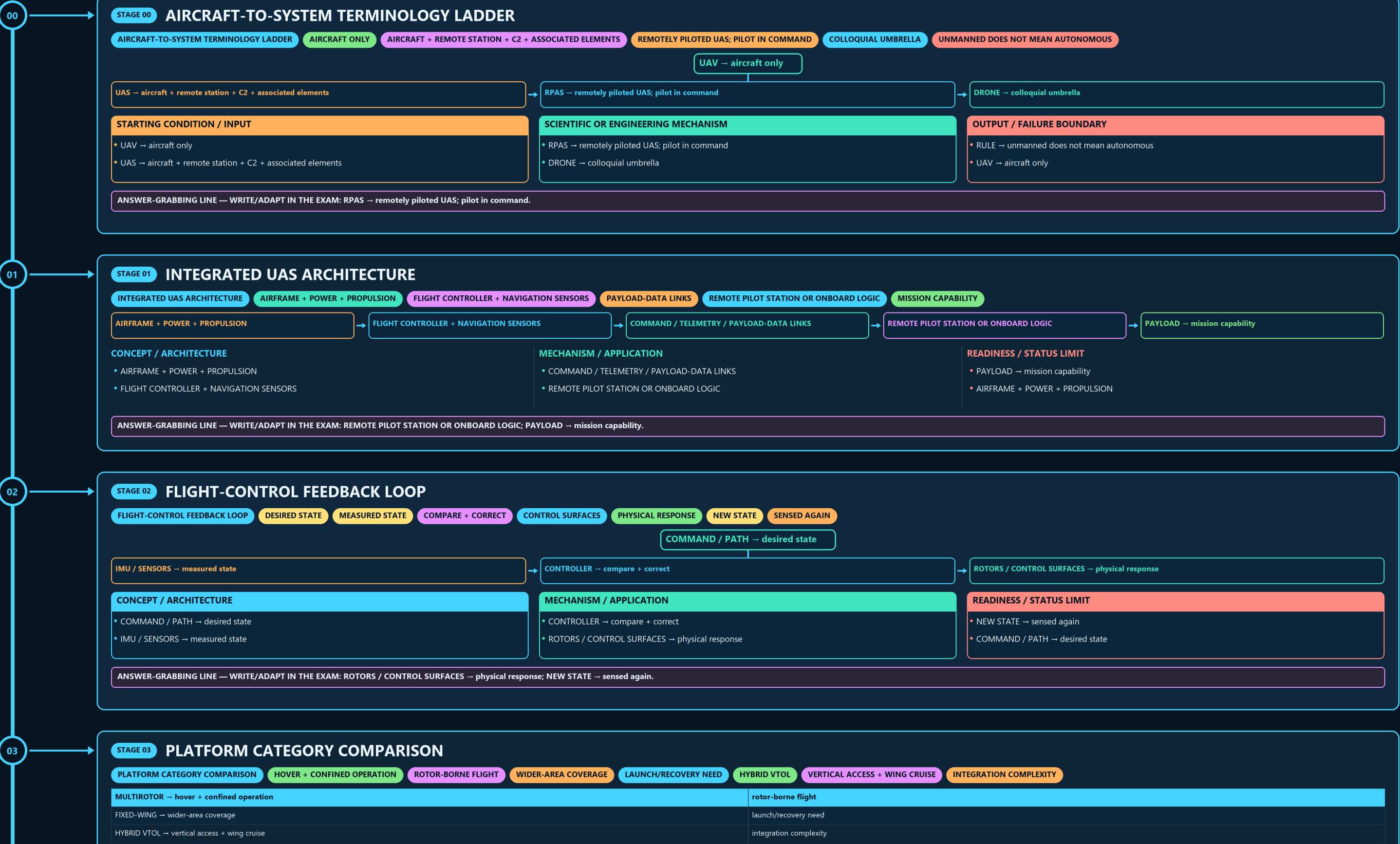
AIRCRAFT-TO-SYSTEM TERMINOLOGY LADDER → INTEGRATED UAS ARCHITECTURE → FLIGHT-CONTROL FEEDBACK LOOP → PLATFORM CATEGORY COMPARISON → HUMAN-CONTROL CONTINUUM → SWARM COORDINATION MESH

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



CONCEPT / ARCHITECTURE

- COMMAND / PATH → desired state
- IMU / SENSORS → measured state

MECHANISM / APPLICATION

- CONTROLLER → compare + correct
- ROTORS / CONTROL SURFACES → physical response

READINESS / STATUS LIMIT

- NEW STATE → sensed again
- COMMAND / PATH → desired state

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROTORS / CONTROL SURFACES → physical response; NEW STATE → sensed again.

03

STAGE 03

PLATFORM CATEGORY COMPARISON

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HOVER + CONFINED OPERATION

ROTOR-BORNE FLIGHT

WIDER-AREA COVERAGE

LAUNCH/RECOVERY NEED

HYBRID VTOL

VERTICAL ACCESS + WING CRUISE

INTEGRATION COMPLEXITY

MULTIROTOR → hover + confined operation	rotor-borne flight
FIXED-WING → wider-area coverage	launch/recovery need
HYBRID VTOL → vertical access + wing cruise	integration complexity

SYSTEM / DEVICE / INSTITUTION

- MULTIROTOR → hover + confined operation
- rotor-borne flight
- FIXED-WING → wider-area coverage

DECISIVE TECHNICAL DISTINCTION

- launch/recovery need
- HYBRID VTOL → vertical access + wing cruise
- integration complexity

STATUS / UNIT / EVIDENCE LIMIT

- PAYLOAD CHANGES → mass + power + data + control burden
- NO LABEL → universal range/endurance guarantee

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PAYLOAD CHANGES → mass + power + data + control burden.

04

STAGE 04

HUMAN-CONTROL CONTINUUM

HUMAN-CONTROL CONTINUUM

HUMAN DIRECTLY COMMANDS

FIXED SEQUENCE

SENSES AND SELECTS ACTIONS

AI SUBSYSTEM

PERCEPTION/DECISION SUPPORT, NOT LEGAL AUTHORITY

HUMAN OVERSIGHT + DEFINED FALLBACK

CONCEPT / ARCHITECTURE

- TELEOPERATION → human directly commands
- AUTOMATION → fixed sequence

MECHANISM / APPLICATION

- AUTONOMY → senses and selects actions
- AI SUBSYSTEM → perception/decision support, not legal authority

READINESS / STATUS LIMIT

- ACCOUNTABILITY → human oversight + defined fallback
- TELEOPERATION → human directly commands

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AI SUBSYSTEM → perception/decision support, not legal authority.

05

STAGE 05

SWARM COORDINATION MESH

SWARM COORDINATION MESH

MULTIPLE AIRCRAFT

SELF AND NEIGHBOUR STATE

SHARED TASK/STATE INFORMATION

CONTROL RULES

ALLOCATION + FORMATION + ADAPTATION

FAILURE TEST

LINK LOSS OR MEMBER LOSS

MEMBERS → multiple aircraft

→

LOCALISATION → self and neighbour state

→

COMMUNICATION → shared task/state information

→

CONTROL RULES → allocation + formation + adaptation

→

FAILURE TEST → link loss or member loss

CONCEPT / ARCHITECTURE

- MEMBERS → multiple aircraft
- LOCALISATION → self and neighbour state

MECHANISM / APPLICATION

- COMMUNICATION → shared task/state information
- CONTROL RULES → allocation + formation + adaptation

READINESS / STATUS LIMIT

- FAILURE TEST → link loss or member loss
- MEMBERS → multiple aircraft

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAILURE TEST → link loss or member loss.

06

STAGE 06

VLOS TO BVLOS ASSURANCE BRIDGE

VLOS TO BVLOS ASSURANCE BRIDGE

PILOT MAINTAINS UNAIDED VISUAL CONTACT

AIRCRAFT BEYOND VISUAL ENVELOPE

RELIABLE C2 + NAVIGATION + DETECT-AND-AVOID

LOST LINK + DIVERSION + TERMINATION LOGIC

TRIAL AUTHORISATION

ROUTINE PERMISSION

VLOS → pilot maintains unaided visual contact

→

BVLOS → aircraft beyond visual envelope

→

BRIDGE → reliable C2 + navigation + detect-and-avoid

→

CONTINGENCY → lost link + diversion + termination logic

CONCEPT / ARCHITECTURE

- VLOS → pilot maintains unaided visual contact
- BVLOS → aircraft beyond visual envelope

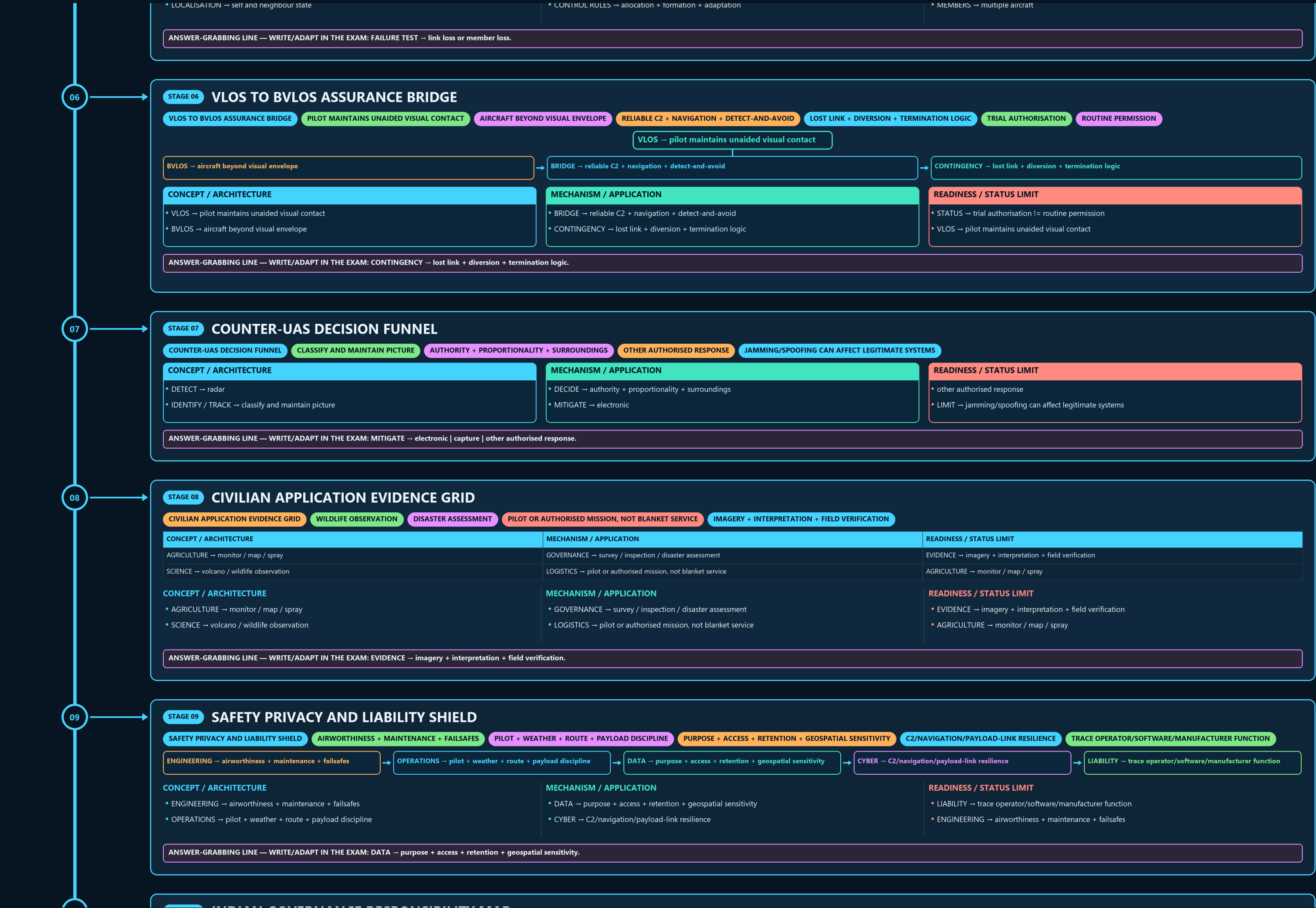
MECHANISM / APPLICATION

- BRIDGE → reliable C2 + navigation + detect-and-avoid
- CONTINGENCY → lost link + diversion + termination logic

READINESS / STATUS LIMIT

- STATUS → trial authorisation != routine permission
- VLOS → pilot maintains unaided visual contact

DRONES, UAVS AND ROBOTICS POLICY • TILE 2/4 • same master rows 2359-5593 of 10310 • continues below



09

STAGE 09 SAFETY PRIVACY AND LIABILITY SHIELD

SAFETY PRIVACY AND LIABILITY SHIELD AIRWORTHINESS + MAINTENANCE + FAILSAFES PILOT + WEATHER + ROUTE + PAYLOAD DISCIPLINE PURPOSE + ACCESS + RETENTION + GEOSPATIAL SENSITIVITY C2/NAVIGATION/PAYLOAD-LINK RESILIENCE TRACE OPERATOR/SOFTWARE/MANUFACTURER FUNCTION

ENGINEERING → airworthiness + maintenance + failsafes → OPERATIONS → pilot + weather + route + payload discipline → DATA → purpose + access + retention + geospatial sensitivity → CYBER → C2/navigation/payload-link resilience → LIABILITY → trace operator/software/manufacturer function

CONCEPT / ARCHITECTURE

- ENGINEERING → airworthiness + maintenance + failsafes
- OPERATIONS → pilot + weather + route + payload discipline

MECHANISM / APPLICATION

- DATA → purpose + access + retention + geospatial sensitivity
- CYBER → C2/navigation/payload-link resilience

READINESS / STATUS LIMIT

- LIABILITY → trace operator/software/manufacturer function
- ENGINEERING → airworthiness + maintenance + failsafes

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DATA → purpose + access + retention + geospatial sensitivity.

10

STAGE 10 INDIAN GOVERNANCE RESPONSIBILITY MAP

INDIAN GOVERNANCE RESPONSIBILITY MAP CIVIL REGULATION AIRSPACE AND TRAFFIC SERVICES AVIATION SECURITY DIGITAL SKY WORKFLOW PLATFORM, NOT ATC

DGCA → civil regulation / certification → AAI / ATC → airspace and traffic services → BCAS → aviation security

MOCA → policy / notification

CONCEPT / ARCHITECTURE

- MOCA → policy / notification
- DGCA → civil regulation / certification

MECHANISM / APPLICATION

- AAI / ATC → airspace and traffic services
- BCAS → aviation security

READINESS / STATUS LIMIT

- DIGITAL SKY → workflow platform, not ATC
- MOCA → policy / notification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AAI / ATC → airspace and traffic services.

11

STAGE 11 APPROVAL-TO-OUTCOME STATUS RAIL

APPROVAL-TO-OUTCOME STATUS RAIL LEGAL TEXT TYPE CERTIFICATE DISTINCT COMPLIANCE CONDITIONS AUTHORISED OR SUPPORTED ROUTE NOT OPERATIONAL DEPLOYMENT NOT VERIFIED SAFETY, SCALE OR OUTCOME

RULE / AMENDMENT → legal text → TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions → TRIAL / SCHEME / INCENTIVE → authorised or supported route → PROCUREMENT / DELIVERY → not operational deployment → DEPLOYMENT → not verified safety, scale or outcome

CONCEPT / ARCHITECTURE

- RULE / AMENDMENT → legal text
- TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions

MECHANISM / APPLICATION

- TRIAL / SCHEME / INCENTIVE → authorised or supported route
- PROCUREMENT / DELIVERY → not operational deployment

READINESS / STATUS LIMIT

- DEPLOYMENT → not verified safety, scale or outcome
- RULE / AMENDMENT → legal text

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DEPLOYMENT → not verified safety, scale or outcome.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT THE MOST IMPORTANT ANALYTICAL SPLIT IS BETWEEN AIRFRAME MANUFACTURING THE LATTER DEPENDS ON SENSORS, SOFTWARE, COMMUNICATIONS AND PAYLOAD DRONE POLICY SITS AT THE INTERSECTION OF AVIATION LAW

ROBOTICS CAPABILITY IS INCREASINGLY DUAL-USE AGRICULTURE SPRAYING AND LOGISTICS CAN COEXIST WITH DEFENCE-SUPPORT OR RULE-MAKING AND SCHEME-LEVEL ANCHOR. COMPLIANCE AND CIVIL-DRONE REGULATORY AUTHORITY.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The most important analytical split is between airframe manufacturing and mission capability
- the latter depends on sensors, software, communications and payload integration.
- CAUTION: Drone policy sits at the intersection of aviation law, digital governance, privacy, internal security and industrial policy.
- CAUTION: Robotics capability is increasingly dual-use: agriculture spraying and logistics can coexist with defence-support or surveillance relevance.

READINESS, UNIT OR STATUS LIMIT

- MoCA: rule-making and scheme-level anchor.
- DGCA: compliance and civil-drone regulatory authority.
- Digital Sky: public-facing airspace and permission architecture.
- PLI for drones and drone components: industrial push for indigenous ecosystem capability.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Training/certification ecosystem: necessary for safe scaling of operations.
- Agriculture: precision application and monitoring can reduce time and improve coverage.
- Disaster response: rapid imaging and access to difficult terrain improves first-response intelligence.
- Surveying/logistics/infrastructure inspection: drones reduce time and increase reach.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.